📌 BARBERBOOK APP - BRIEF ALGORITHM

1️⃣ AUTHENTICATION FLOW

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1. Splash Screen → Check login status
   ├─ If logged in → Fetch role → Go to Dashboard
   └─ If not logged in → Login Screen

2. Login: Enter phone → Send OTP → Auto-read via SMS API → Verify

3. New User? → Select Role (Customer/Barber)
   ├─ Customer: Name, Email, Location → Save
   └─ Barber: Name, Shop, Address, Photo, Services, Hours → Save

4. Navigate to respective dashboard
```

2️⃣ CUSTOMER SIDE

text

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📍 BROWSE BARBERS
Get GPS location → Query nearby barbers → Calculate distance → Sort by nearest → Display with ratings & availability

📅 BOOK APPOINTMENT
Select barber → Choose services (auto-calc price & duration) → Pick date → Check availability → Select time slot → Verify no double booking → Save to Firestore → Send notification to barber

📝 MANAGE BOOKINGS
View all bookings (upcoming/past) → Click Cancel → Confirm → Update status → Release slot → Notify barber
Click Reschedule → Pick new date/time → Check availability → Update booking

⭐ REVIEW
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After completed appointment → Rate 1-5 stars → Write review → Save → Update barber's average rating

❤️ FAVORITES

Tap heart icon → Save barber to favorites list → Quick access for re-booking

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3️⃣ BARBER SIDE

text

📊 DASHBOARD

Load today's bookings → Display list with customer, time, services, amount → Show summary (total appointments, earnings) → Real-time updates

✂️ MANAGE SERVICES

View all services → Add/Edit/Delete → Toggle enabled/disabled → Customers see only enabled services

🗓️ SCHEDULE

Set weekly working hours (day-wise start/end time) → Block specific dates (vacation/holiday) → Toggle Online/Offline status

💰 EARNINGS

View earnings (daily/weekly/monthly) → See total bookings + average value → Display chart/graph

✅ APPOINTMENTS

Mark as Done → Change status to completed → Add to earnings

Cancel → Select reason → Notify customer → Release slot

4️⃣ NOTIFICATION SYSTEM (FCM)

Event	Trigger	Recipient
New Booking	Customer books	Barber

Event	Trigger	Recipient
Booking Confirmed	Barber accepts	Customer
Cancellation	Either party cancels	Other party
Reminder	1 day before appointment	Both
Review Request	24 hrs after completion	Customer

5️⃣ DATA STRUCTURE

text

users { id, name, phone, role }

barbers { id, name, shop, location, avgRating, isOnline, schedule, blockedDates }

services { id, barberId, name, price, duration, isEnabled }

bookings { id, customerId, barberId, services[], totalPrice, date, timeSlot, status }

reviews { id, bookingId, barberId, rating, review }

favorites { customerId, barberId }

6️⃣ BOOKING STATE MACHINE

text

PENDING → CONFIRMED → COMPLETED

↓ ↓

CANCELLED ← CANCELLED

7️⃣ TECH STACK SNAPSHOT

text

Layer | Technology

———————┼———————

Language | Kotlin

Architecture | MVVM + LiveData

Auth | Firebase Phone Auth

Database | Firebase Firestore

Maps | Google Maps SDK

Notifications | FCM

UI | XML + Material 3

8️⃣ HIGH-LEVEL FLOW

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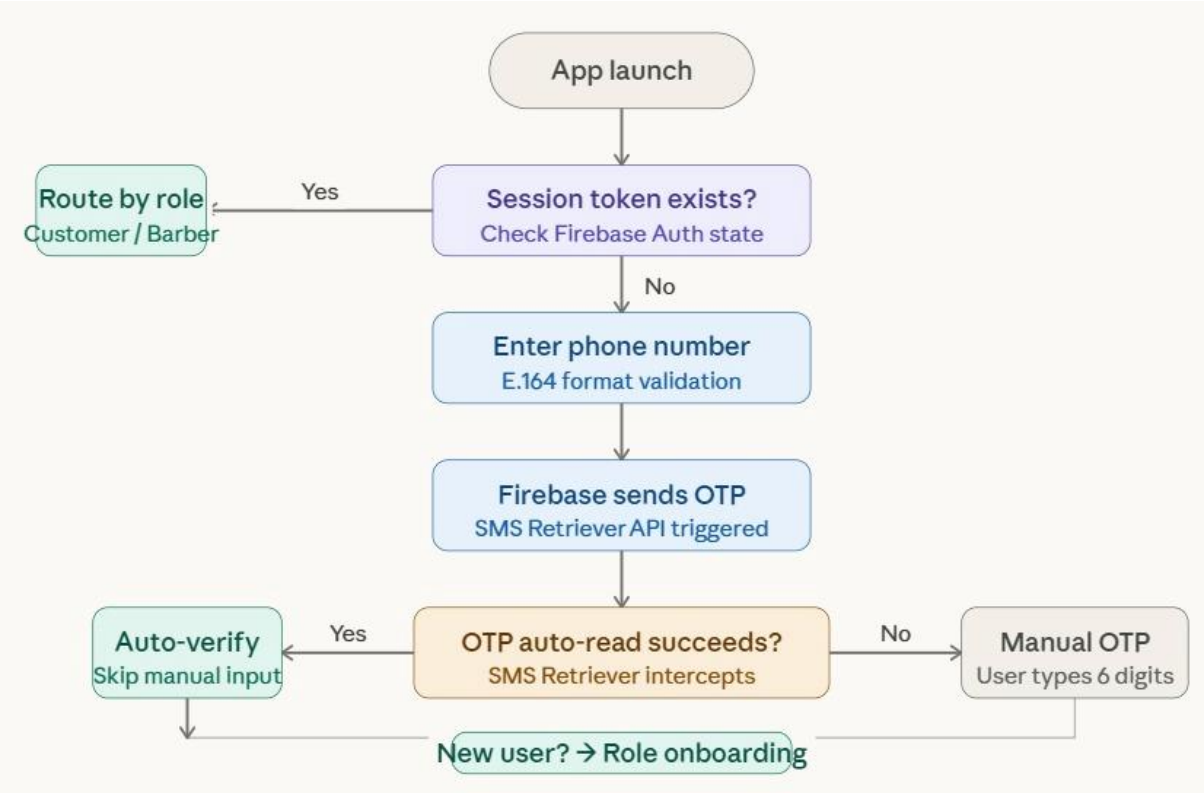
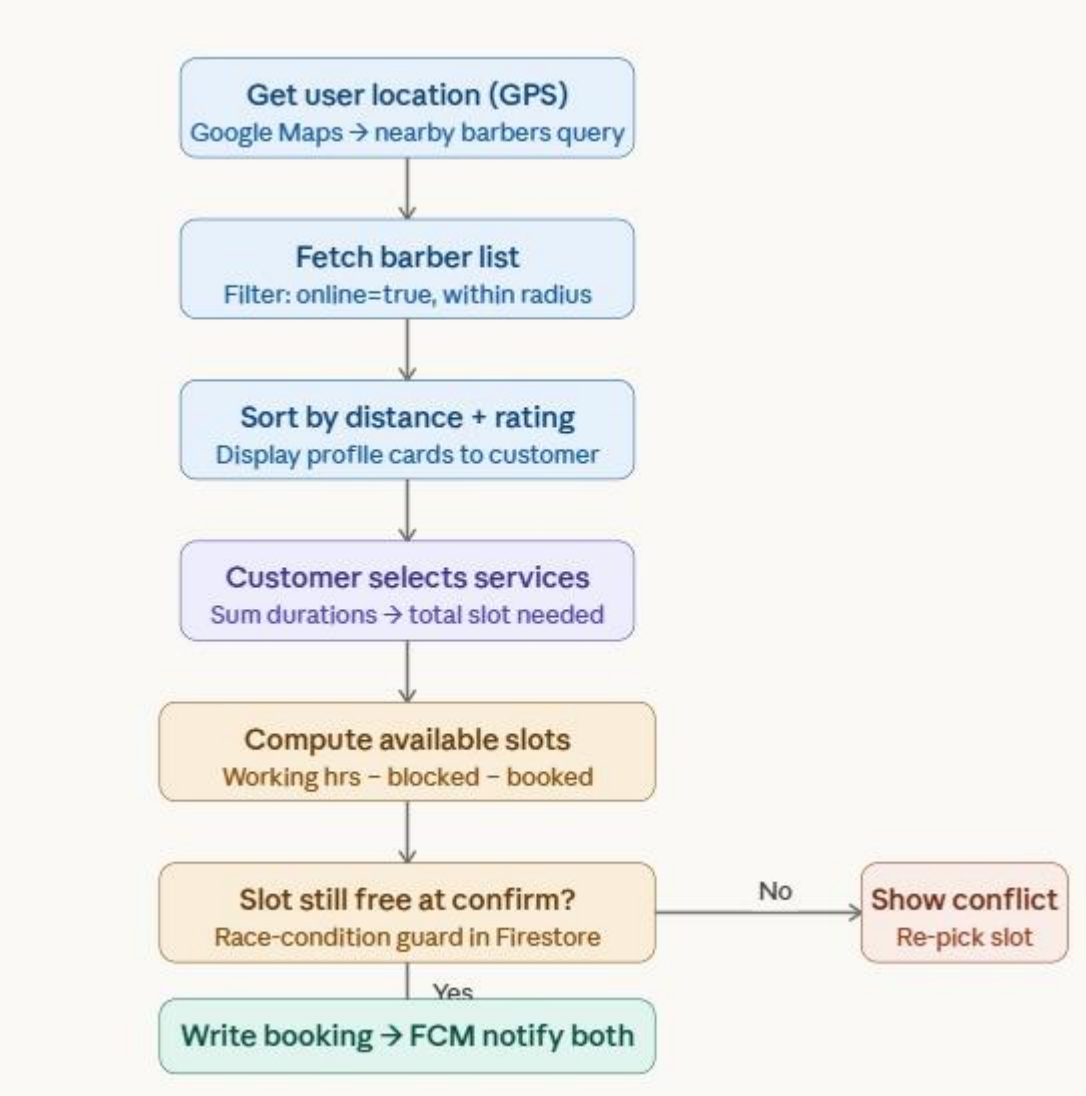
```
START
  ↓
LOGIN (Phone + OTP)
  ↓
ROLE SELECTION
  ↓
┌──────────────────┐
│  CUSTOMER   |   BARBER
│ • Browse    | • Dashboard
│ • Book      | • Manage Services
│ • Manage    | • Set Schedule
│ • Review    | • View Earnings
│ • Favorites | • Handle Appointments
└──────────────────┘
  ↓
LOGOUT / EXIT
```

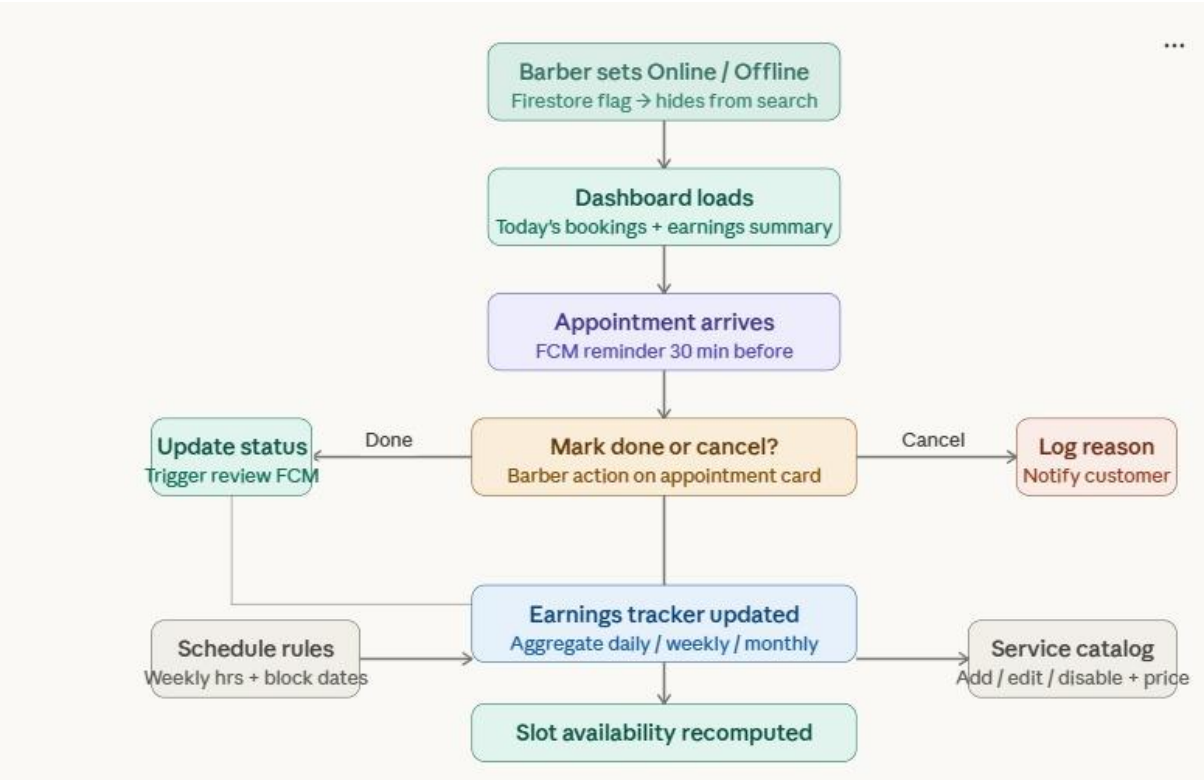
🔑 KEY ALGORITHMS SUMMARY

Feature	Core Logic
Distance Calculation	Haversine formula (latitude/longitude)

Feature	Core Logic
Slot Availability	Working hours - existing bookings - blocked dates
Double Booking Prevention	Check Firestore before saving
Auto OTP Read	SMS Retriever API + BroadcastReceiver
Real-time Updates	Firestore listeners + FCM push
Rating Update	Average all reviews when new review added




Get user location (GPS)
Google Maps → nearby barbers query
Fetch barber list
Filter: online=true, within radius
Sort by distance + rating
Display profile cards to customer
Customer selects services
Sum durations → total slot needed
Compute available slots
Working hrs − blocked − booked
Slot still free at confirm?
Race-condition guard in Firestore
No
Show conflict
Re-pick slot
Yes
Write booking → FCM notify both


Barber sets Online / Offline
Firestore flag → hides from search
Dashboard loads
Today's bookings + earnings summary
Appointment arrives
FCM reminder 30 min before
Mark done or cancel?
Barber action on appointment card
Done
Cancel
Update status
Trigger review FCM
Log reason
Notify customer
Earnings tracker updated
Aggregate daily / weekly / monthly
Schedule rules
Weekly hrs + block dates
Service catalog
Add / edit / disable + price
Slot availability recomputed